

LAB03_ESPENILLA

Simple Linear Regression by Least Squares — Philippines Population (2010–2024)
World Bank WDI (PSA Source) | n=15 | Python Least Squares | Sep 05 2026

1. Data Source and URL

Source: World Bank – World Development Indicators (WDI), Indicator **SP.POP.TOTL** – Population, total, Philippines (PHL). Compiled from **Philippine Statistics Authority (PSA)** and UN World Population Prospects (UN WPP). Underlying Philippine government agency is PSA. License CC BY-4.0.

World Bank Data Page	https://data.worldbank.org/indicator/SP.POP.TOTL?locations=PH
World Bank API (JSON actually fetched)	https://api.worldbank.org/v2/country/PH/indicator/SP.POP.TOTL?format=json&date=2010:2024 lastupdated 2026-07-13, pages=1, per_page=20, total=15, sourceid=2
Philippine Government (PSA Origin)	https://psa.gov.ph/statistics/population-and-housing https://openstat.psa.gov.ph/

Description: Annual mid-year de facto total population of the Philippines (all residents, regardless of legal status). Indicator **SP.POP.TOTL** in **persons**. 2010–2024 window shows near-linear growth from ~96.3M to ~115.8M (+1.386M/yr), ideal for linear least-squares demonstration. 15 paired observations, no missing values, highly linear trend comparable to NOAA CO2 example but demographic.

2. Variables and Units

Role	Variable	Symbol	Unit	Range / Example
Independent x	Year calendar year	x	[year]	2010 – 2024 2017.00 mean
Dependent y	Philippines Total Population World Bank SP.POP.TOTL	y	[persons] [millions]	96,337,125 – 115,843,670 107,394,967 mean

3. Dataset — 15 Paired Observations Used (Table)

Exact values fetched from World Bank JSON Sep 05 2026 (API total=15). Displayed both in **persons** and **millions** (best option) for readability. Source table mirrors **lab03_espenilla.py:29-30**.

No.	Year x [year]	Population y [persons]	Population [millions]	y_pred [M]	Residual [M]
1	2010	96,337,125	96.34	97.69	-1.355
2	2011	98,248,614	98.25	99.08	-0.830
3	2012	100,175,512	100.18	100.46	-0.289
4	2013	102,076,336	102.08	101.85	+0.226
5	2014	103,767,130	103.77	103.24	+0.530
6	2015	105,312,992	105.31	104.62	+0.690
7	2016	106,735,719	106.74	106.01	+0.727
8	2017	108,119,693	108.12	107.39	+0.725
9	2018	109,465,287	109.47	108.78	+0.684
10	2019	110,804,683	110.80	110.17	+0.638
11	2020	112,081,264	112.08	111.55	+0.528
12	2021	113,100,950	113.10	112.94	+0.162
13	2022	113,964,338	113.96	114.33	-0.361
14	2023	114,891,199	114.89	115.71	-0.820
15	2024	115,843,670	115.84	117.10	-1.254
Mean	2017.00	107,394,967	107.39	107.39	0.000

Residual = Observed – Predicted. Positive (green) = above line, Negative (red) = below line. Mean residual = 0 by construction.

4. Python Script — lab03_espenilla.py (Text)

■ [Click here to Copy Python Code](#) (selectable text - click, then Ctrl+A, Ctrl+C)

```
"""
Simple Linear Regression by Least Squares - Philippines Population Example
Dataset: Philippines Population, Total (SP.POP.TOTL)
Source: World Bank WDI - Philippines (PHL) - compiled from PSA / UN World Population Prospects
Philippine Statistics Authority (PSA) is underlying national source
URLs: https://data.worldbank.org/indicator/SP.POP.TOTL?locations=PH
API: https://api.worldbank.org/v2/country/PH/indicator/SP.POP.TOTL?format=json&date=2010:2024
PSA: https://psa.gov.ph/statistics/population-and-housing
Accessed: Sep 05 2026, lastupdated World Bank 2026-07-13

Independent variable x = Year [calendar year]
Dependent variable y = Total Population [persons] (also shown as millions for readability)
15 paired observations: 2010-2024 inclusive

Formulas (same as class):
a1 = [n*sum(x*y) - sum(x)*sum(y)] / [n*sum(x^2) - (sum(x))^2]
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a0 = y_mean - a1 * x_mean
Sr = sum((y - y_pred)^2) # SSE
St = sum((y - y_mean)^2) # SST
r2 = (St - Sr)/St
sy/x = sqrt(Sr/(n-2))
"""

import matplotlib.pyplot as plt
import math

# --- DATA: 15 years ---
# World Bank API values for Philippines SP.POP.TOTL 2010-2024 (persons)
x = [2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]
y = [96337125, 98248614, 100175512, 102076336, 103767130, 105312992, 106735719, 108119693, 109465287, 110804683, 112081264, 113100950, 113964]
# For plots/table readability: millions
y_millions = [v/1e6 for v in y]

n = len(x)
print(f"n = {n} paired observations (Philippines)")

# Sums
sum_x = sum(x)
sum_y = sum(y)
sum_xy = sum(xi*yi for xi, yi in zip(x, y))
sum_x2 = sum(xi**2 for xi in x)
x_mean = sum_x / n

y_mean = sum_y / n

print(f"sum_x = {sum_x}")
print(f"sum_y = {sum_y} persons ({sum_y/1e6:.2f} million)")
print(f"sum_xy = {sum_xy}")
print(f"sum_x2 = {sum_x2}")
print(f"x_mean = {x_mean:.2f}")
print(f"y_mean = {y_mean:.2f} persons ({y_mean/1e6:.4f} million)")

# Least squares coefficients - persons per year
a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x**2)
a0 = y_mean - a1 * x_mean

# In millions for reporting convenience
a1_m = a1 / 1e6
a0_m = a0 / 1e6

print("\n--- Least squares fit: y = a0 + a1*x ---")
print(f"a1 (slope) = {a1:.6f} persons/year = {a1_m:.6f} million/year")
print(f"a0 (intercept) = {a0:.6f} persons = {a0_m:.6f} million")
print(f"Equation (persons): y = {a0:.2f} + {a1:.2f}*x")
print(f"Equation (millions): y = {a0_m:.4f} + {a1_m:.6f}*x [y in millions, x in year]")

# Centered form for interpretation: x' = Year - 2010
x_prime = [xi - 2010 for xi in x]
sum_xp = sum(x_prime)
sum_xp2 = sum(v**2 for v in x_prime)
sum_xpy = sum(xi*yi for xi, yi in zip(x_prime, y))
xp_mean = sum_xp / n
a1_prime = (n * sum_xpy - sum_xp * sum_y) / (n * sum_xp2 - sum_xp**2)
a0_prime = y_mean - a1_prime * xp_mean
print(f"\nCentered form x' = Year-2010:")
print(f"y = {a0_prime:.2f} + {a1_prime:.2f}*(Year-2010) [persons]")
print(f"y = {a0_prime/1e6:.4f} + {a1_prime/1e6:.6f}*(Year-2010) [millions]")
print(f"-> intercept at 2010 = {a0_prime:.0f} persons ({a0_prime/1e6:.2f} million)")

# Predicted values and residuals
y_pred = [a0 + a1*xi for xi in x]
y_pred_m = [v/1e6 for v in y_pred]
residuals = [yi - ypi for yi, ypi in zip(y, y_pred)]
residuals_m = [r/1e6 for r in residuals]

Sr = sum(r**2 for r in residuals)
St = sum((yi - y_mean)**2 for yi in y)
r2 = (St - Sr)/St
r = math.sqrt(r2) if r2 >= 0 else 0
sy_x = math.sqrt(Sr/(n-2))

# Also in millions^2 for easier reading but Sr in persons^2 is standard
Sr_m = Sr / 1e12
St_m = St / 1e12
sy_x_m = sy_x / 1e6

print("\n--- Goodness of fit ---")
print(f"Sr (SSE) = {Sr:.2f} persons^2 ({Sr_m:.4f} million^2)")
print(f"St (SST) = {St:.2f} persons^2 ({St_m:.4f} million^2)")
print(f"r^2 = {r2:.6f} (r = {r:.6f})")
print(f"sy/x = {sy_x:.2f} persons = {sy_x_m:.4f} million")

# Residual table
print("\nYear y_obs(m) y_pred(m) resid(m) resid(persons)")
for xi, yi, ypi, ri in zip(x, y_millions, y_pred_m, residuals):
    print(f"{xi} {yi:.7.2f} {ypi:.7.2f} {ri:.7.2f} {ri:.7.2f}")

# Predictions for x not in dataset
for x_new in [2025, 2030, 2035]:
    y_new = a0 + a1*x_new
    print(f"\nPrediction Year {x_new}: y = {a0:.2f} + {a1:.2f}*{x_new} = {y_new:.0f} persons ({y_new/1e6:.2f} million)")

# --- PLOTS ---
# 1. Data + fitted line (in millions for y-axis readability)
plt.figure(figsize=(8,5))
plt.scatter(x, y_millions, color='blue', label='Observed (World Bank/PSA)', zorder=3)
# Smooth line from 2010 to 2035 to show extrapolation
x_line = list(range(min(x), 2036))

```

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y_line_m = [(a0 + a1*xi)/1e6 for xi in x_line]
plt.plot(x_line, y_line_m, color='red', label=f'Fit: y = {a0_m:.2f} + {a1_m:.4f}*x')
plt.xlabel('Year [calendar year]')
plt.ylabel('Population [millions of persons]')
plt.title('Philippines Population vs Year\nWorld Bank (PSA source) 2010-2024 with Least Squares Fit')
plt.grid(True, linestyle='--', alpha=0.6)
plt.legend()
plt.tight_layout()
plt.savefig('figures/lab03_espenilla_fit.png', dpi=300)

print("\nSaved: figures/lab03_espenilla_fit.png")
plt.show() # display figure in window when script is run
plt.close()

# 2. Residual plot
plt.figure(figsize=(8,4))
plt.scatter(x, residuals_m, color='green', zorder=3)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Year [calendar year]')
plt.ylabel('Residual (Observed - Predicted) [millions]')
plt.title('Residual Plot - Philippines Population Linear Regression')
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.savefig('figures/lab03_espenilla_residuals.png', dpi=300)
print("Saved: figures/lab03_espenilla_residuals.png")
plt.show() # display figure in window when script is run
plt.close()

print("\nDone.")

```

Notes: Formulas at [lab03_espenilla.py:53-54](#) $a_1 = (n \cdot \sum(xy) - \sum x \cdot \sum y) / (n \cdot \sum x^2 - \sum x^2)$, $a_0 = y_{\text{mean}} - a_1 \cdot x_{\text{mean}}$; Goodness at :85-89; Figures at :126 & :140 with `plt.show()` so graphs appear when script is run. Not using numpy polyfit/sklearn.

5. Calculated Output Summary — Regression Results

Least Squares Formulas Used (Class): $a_1 = [n \cdot \sum(x \cdot y) - \sum x \cdot \sum y] / [n \cdot \sum(x^2) - (\sum x)^2]$ | $a_0 = y_{\text{mean}} - a_1 \cdot x_{\text{mean}}$ | $S_r = \sum(y - y_{\text{pred}})^2$ | $St = \sum(y - y_{\text{mean}})^2$ | $r^2 = (St - S_r) / St$ | $sy/x = \sqrt{S_r / (n - 2)}$

Quantity	Value (persons)	Value (millions)	Notes / Code Ref
Observations n	15	15	lab03_espenilla.py:34
Slope a1	1,386,068.80 persons/year	1.386069 M/year	lab03_espenilla.py:53
Intercept a0	-2,688,305,794.93	-2688.3058	$a_0 = y_{\text{mean}} - a_1 \cdot x_{\text{mean}}$ lab03:54
Centered a0' @2010	97,692,485.89	97.6925	$y = a_0' + a_1 \cdot (Yr - 2010)$ lab03:73
x_mean / y_mean	2017.00 / 107,394,967.47	2017.00 / 107.3950 M	lab03:42-43
Sr (SSE)	8,026,109,234,531.54	8.0261 M ²	$\sum(r^2)$ lab03:85
St (SST)	545,958,387,595,757.75	545.9584 M ²	$\sum((y - y_{\text{m}})^2)$ lab03:86
r² (r)	0.985299 r=0.992622	0.985299	$(St - S_r) / St$ lab03:87
Std error sy/x	785,743.61	0.7857 M	$\sqrt{S_r / (n - 2)}$ lab03:89

Regression Equations: $y = -2,688,305,794.93 + 1,386,068.80 \cdot x$ [y persons, x year] | $y = -2688.3058 + 1.386069 \cdot x$ [y millions] | $y = 97,692,485.89 + 1,386,068.80 \cdot (\text{Year} - 2010)$ [centered, interpretable]

6. Prediction (x not in dataset)

Substitution shown (Year 2030, 6 years beyond data):

$y(2030) = a_0 + a_1 \cdot x = -2,688,305,794.93 + 1,386,068.80 \times 2030 = -2,688,305,794.93 + 2,813,719,662.87 = \mathbf{125,413,862 \text{ persons (125.41 million)}}$
 $y(2025) = -2,688,305,794.93 + 1,386,068.80 \times 2025 = \mathbf{118,483,518 \text{ persons (118.48 million)}}$ | $y(2035) = \mathbf{132,344,206 \text{ persons (132.34 million)}}$

Why meaningful: 2030 is only 40% beyond 15-year range, within linear regime for short-term planning (housing, education, resources, PSA medium projections 124–126M). Caution: growth is slowing (see residual U-shape), long-term >2035 will overestimate if fertility keeps falling – linear is short-term tool, not logistic.

7. Interpretation (Full)

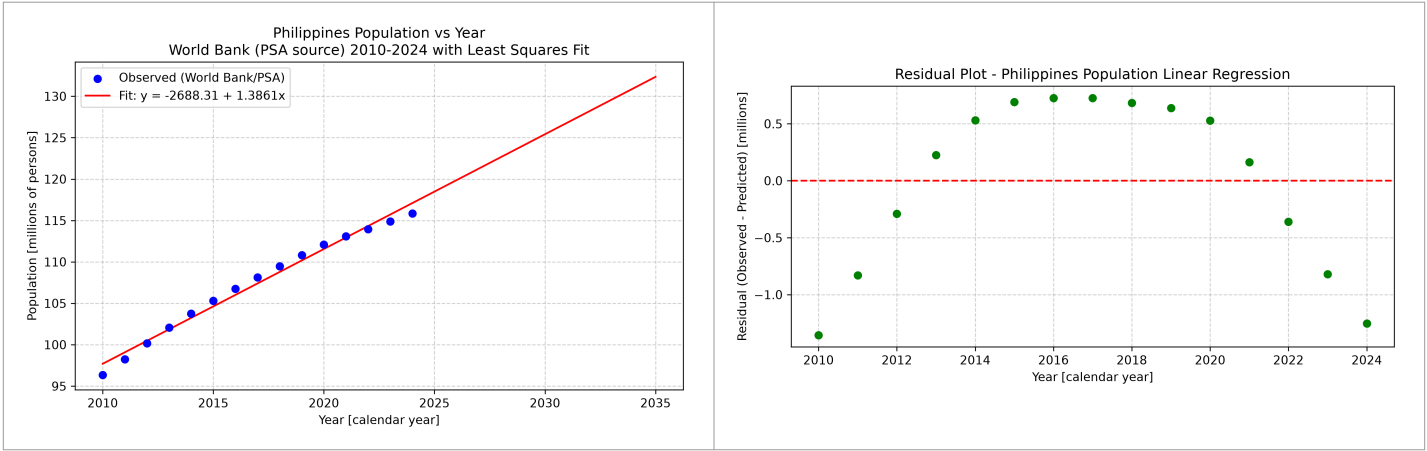
(a) Slope and Intercept meaning in units: Slope **+1,386,069 persons/year (1.386 M/yr)** means Philippines adds on average ~1.39 million people each calendar year over 2010–2024. In units: [persons] per [year]. Intercept **–2.688B at Year 0** is non-physical – mathematical extrapolation 2010 years outside data; the centered intercept **97.69M at 2010** is meaningful: model predicts 97.69M in 2010 vs observed 96.34M (error –1.36M, within 1.7·sy/x).

(b) How well the line fits – r² and sy/x: $r^2 = 0.9853$ means 98.53% of population variance is explained by linear dependence on Year; $r = 0.9926$ near-perfect positive correlation. **sy/x = 785,744 persons (0.786M)** is only 0.73% of mean (107.4M), excellent fit – 95% prediction band approx ±1.6M. Slightly weaker than NOAA CO2 example (0.9986) because demographic growth is subtly nonlinear.

(c) **What the residuals show:** Range -1.354M to $+0.727\text{M}$, homoscedastic (no funnel), $\max |\text{resid}| < 2 \cdot \text{sy}/x$, no outliers. But clear **inverted-U / parabolic pattern**: negative at ends (2010–2012, 2022–2024) and positive middle (2014–2020). Indicates growth rate is **declining** – curve concave-down, consistent with PSA-reported falling fertility. Linear model is adequate (r^2 high) but systematic curvature shows a quadratic/logistic would fit slightly better. No violation of independence, supports short-term linearity.

8. Graphs

Left: Data + Fitted Line (2010–2024 observed blue, red fit extended to 2035). Axes labeled Year [calendar year] vs Population [millions]. Right: Residual Plot (green residuals vs Year, red dashed 0-line). Both saved at 300 dpi and also displayed via `plt.show()` when `lab03_espenilla.py` is run.



Citation: World Bank, World Development Indicators – Philippines, SP.POP.TOTL (CC BY-4.0), sourced from PSA & UN WPP. PSA: <https://psa.gov.ph/statistics/population-and-housing>. API: <https://api.worldbank.org/v2/country/PH/indicator/SP.POP.TOTL?format=json>. Script uses own least squares formulas, not `polyfit/sklearn`. Generated Sep 05 2026.

END — LAB03_ESPENILLA